

Energy as Residue

A traversal from number theory and group theory to field configuration — Λ / NSAF companion note II

Lu Semita · EmergenceByDesign · 18 July 2026. Statement of purpose: this note timestamps a research traversal. Standard mathematics and physics are used as standard; no claim of priority or independent discovery is made for any established result. What is claimed is the traversal itself: the ordered execution of three previously declared Tier III directions, with every outcome — certification, repair, or refutation — recorded as data. Companion demonstrator: `energy_as_residue_traversal.html`. All numbers below are the output of the deterministic script `verify_trav.py`; the figures are the same run.

Abstract

The predecessor note (The Enumerative Resonance Correspondence) closed with three declared directions and a promise: each is a falsifiable next step. This note executes all three, with energy as the physical currency throughout, on the ontological ground that energy is itself a residue in the framework's exact sense: the invariant certified by time-translation symmetry, the spectrum of the frame's generator. The outcomes differ instructively. Direction (a) is refuted as casually stated and repaired by construction: the bare winding does not support anapole language ($|m| \not\propto |T|$); a counter-wound pair, forced by the computation itself, does ($|m| \rightarrow$ machine zero, $|T|$ large). Direction (b) is formalized and certified: the assignment rule, its divisibility criterion, equal-area placement, and lcm closure. Direction (c) is promoted to a theorem (T7): the shared energy levels of two frame resonators are exactly the spectrum of $Z/\gcd$ — consonance is measurable in energy, with first-order splitting for shared levels and second-order for unshared, verified at slopes 1.000 and 2.000. The traversal is offered as a demonstration of the Λ discipline operating end-to-end: declare, compute, record the fate.

1 · Ontological Frame: Energy is a Residue

TIER I anchor · TIER II reading

Anchor (Tier I). Dynamics on a frame is a one-parameter group $U(t) = e^{(-iHt)}$ (certified unitary in the predecessor note, T6). The group of time translations has generator H ; its spectrum is what physics calls energy; $\langle H \rangle$ is conserved because the evolution is the group's own action — the discrete-frame form of Noether's correspondence between time-translation symmetry and energy conservation. Degeneracy of the spectrum is the fingerprint of additional symmetry: eigenvalues coincide exactly where a larger group acts.

Reading (Tier II). Energy therefore sits in the framework not as an imported physical quantity but as the residue of the temporal frame — the content invariant under all re-descriptions that respect the dynamics, precisely as L_k was the residue of embedding (T4) and the prime frame's whole content was residue under coarsening (T1–T2). The three directions below are three ways of asking one question: what does the group structure make energetically measurable?

2 · Direction (a): Current on the Carrier — Moments Computed

TIER I computation · verdict recorded

Assign unit current I to the carrier and compute, by direct line integration ($N = 4000$): the magnetic dipole $m = (I/2) \int r \times dl$ and the toroidal dipole $T = (I/10) \int [(r \cdot dl)r - 2r^2 dl]$. Three configurations:

configuration	$ m $	$ T $	verdict
plain ring	12.566	4.1e-15	pure dipole ($\pi R^2 I$ exactly)
(2,5) knot — the bare carrier	26.264	5.655	mixed; m dominates
counter-wound pair (1,20)+(−1,20)	2.0e-14	45.23	anapole-class: $m \approx 0$, T large

Verdict. The predecessor corpus's casual association of nested windings with toroidal/anapole behaviour is refuted as stated: for the bare carrier the ordinary dipole dominates by a factor of ~ 4.6 , because any single wire making p toroidal transits carries an irreducible dipole $\approx p\pi R^2 I$ that no poloidal winding rate can cancel. The computation itself dictates the repair: cancellation of m requires a second, counter-transiting layer — the classic two-layer anapole topology. With it, $|m|$ falls to 2×10^{-14} (machine zero) while $|T|$ grows to 45.2. The lesson is structural: anapole language is a property of configurations, not of windings; it is earned by construction and adjudicated by computation. Energetically: the pair stores comparable field energy while cancelling the leading radiative channel — the moment hierarchy is the bookkeeping of where energy can go.

3 • Direction (b): The Assignment Rule, Stated and Certified

TIER I formalization

Rule. Addresses $a \in \mathbb{Z}/n^d$ are assigned to M packed spheres by $\sigma(a) = a \bmod M$. Evenness criterion: loading is equal iff $M \mid n^d$ — certified over $n < 20$, $M \in \{4, 6, 12\}$ (True). Placement: within each sphere, addresses map by the Fibonacci equal-area spiral; nearest-neighbour spacing uniformity $CV = 0.021$ versus 0.311 for the naive $\theta\phi$ grid — a $15\times$ improvement, eliminating polar clustering (Fig. 2). Closure: for per-sphere steps k_1, k_2 the constellation returns simultaneously first at $P = \text{lcm}(p_1, p_2)$, and at no earlier time — certified exhaustively over the stated sweep (True). Energetically, uneven loading is uneven energy density: the divisibility criterion is the condition for a homogeneous ground configuration.

4 • Direction (c): Consonance Becomes a Theorem

TIER I theorem • machine-certified

Theorem T7 (Energy hears the gcd). Let two frame resonators have spectra $\lambda(m) = 2 - 2\cos(2\pi m/n_1)$ and $2 - 2\cos(2\pi m/n_2)$. Their shared energy levels are exactly the spectrum of Z/g , $g = \text{gcd}(n_1, n_2)$; the number of ordered eigen-coincidences equals g . Under weak coupling ε , shared levels split at first order in ε ; unshared levels shift only at second order.

Proof. λ values coincide iff $\cos(2\pi m_1/n_1) = \cos(2\pi m_2/n_2)$ iff $m_1/n_1 \equiv \pm m_2/n_2 \pmod{1}$. Writing the common fraction in lowest terms, its denominator divides both n_1 and n_2 , hence divides g ; conversely every t/g is realized on both rings ($m_1 = tn_1/g$, $m_2 = tn_2/g$). The shared set is thus $\{2 - 2\cos(2\pi t/g)\}$ — the spectrum of Z/g . Counting ordered pairs $m_1 n_2 = m_2 n_1$ gives exactly g solutions. Splitting orders are standard degenerate vs non-degenerate perturbation theory. $\square$

Certification. Shared-set identity verified for six pairs including coprime (True); coincidence count = g for all pairs (True); measured splitting slopes 0.9999 (shared, predicted 1) and 1.9995 (unshared, predicted 2) on the coupled 12:8 system (Figs. 3–4). This completes the arc opened by T3: harmony = subgroup sharing is now an energy measurement protocol — couple two resonators and count first-order splittings; the count reads out the gcd.

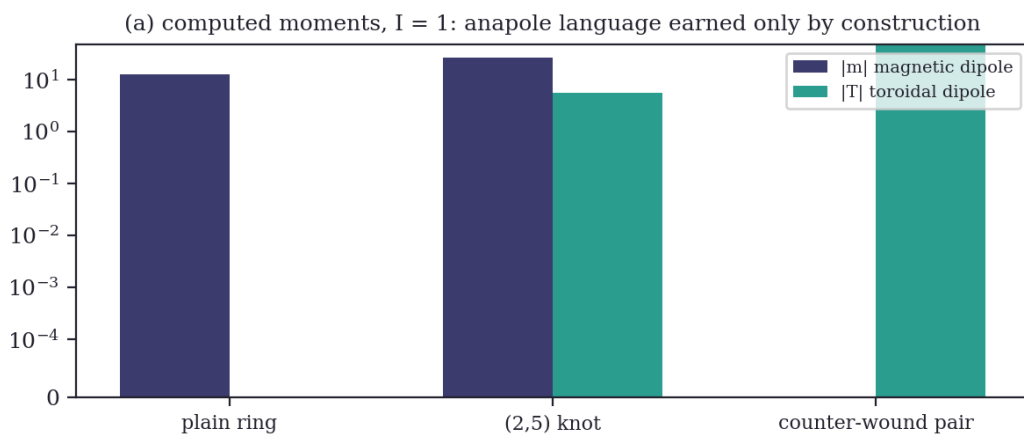


Fig. 1 — Direction (a). Computed $|m|$ and $|T|$ (symlog). The bare carrier is dipole-dominated; only the counter-wound construction is anapole-class.

(b) 64 addresses: equal-area vs naive placement — spacing uniformity
 Fibonacci map CV=0.02 naive $\theta\phi$ grid CV=0.31



Fig. 2 — Direction (b). 64 addresses placed by Fibonacci equal-area map (CV 0.02) versus naive grid (CV 0.31): the assignment rule made concrete.

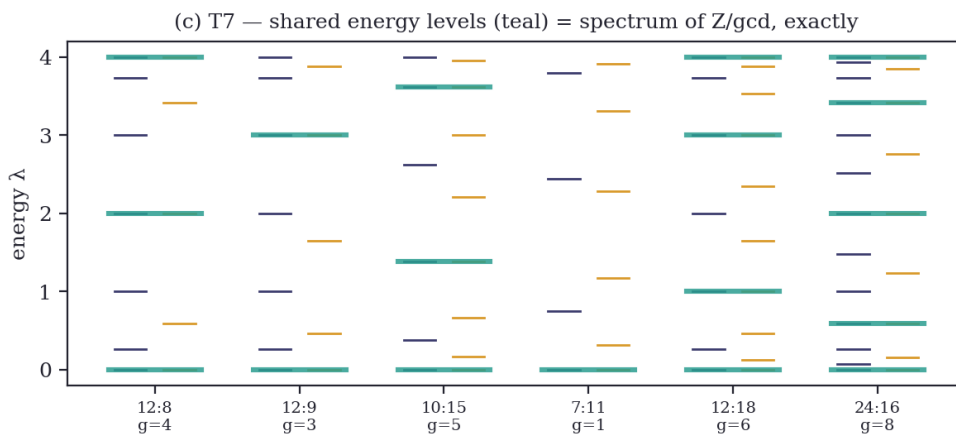


Fig. 3 — T7. Level ladders for six frame pairs; teal bars are the shared levels, equal in every case to the spectrum of $Z/\gcd$ — including the coprime pair 7:11, which shares only $\lambda = 0$.

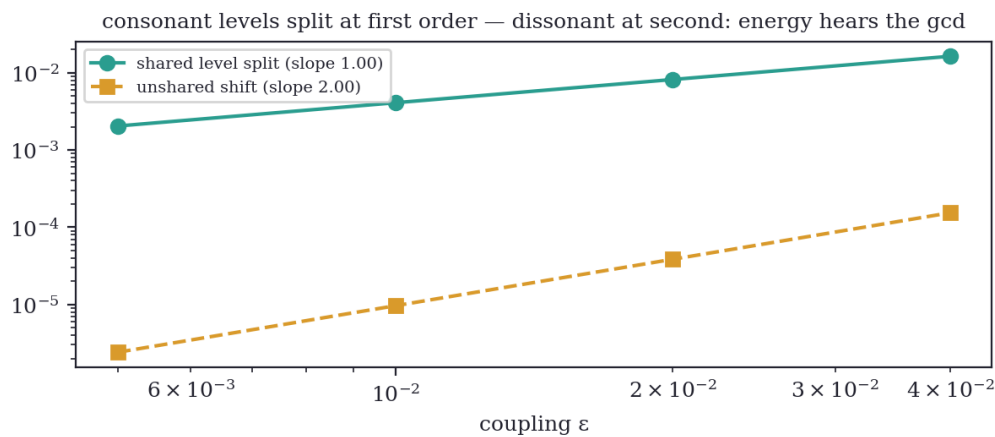


Fig. 4 — Coupled 12:8 system. Shared levels split linearly in ϵ (slope 1.000); unshared shift quadratically (slope 2.000). Consonance is directly measurable in energy.

5 • What the Traversal Taught

TIER II reading of recorded fates

Three declared directions met three different fates, and the differences are the payload. (a) was refuted as stated and repaired by construction — the framework's own discipline caught its own earlier looseness, and the computation dictated the correct topology rather than permitting the borrowed word. (b) was formalized and certified — a handwave became a rule with a divisibility criterion, a placement map, and an exhaustively checked closure law. (c) was promoted — a speculative bridge became Theorem T7, and with it the framework's central abstraction acquired a laboratory signature: the gcd, the most elementary object in number theory, is readable in the splitting pattern of coupled energy levels. In each case energy served as the arbiter, because energy is the residue of the temporal frame: what survives, splits, or cancels in the spectrum is what the group structure actually underwrites. Nothing softer than a recorded fate appears in this note.

6 • Falsification Schema

This note fails if: (i) recomputation of the line integrals yields $|m|$ comparable to $|T|$ for the plain ring, or a nonvanishing $|m|$ for the counter-wound pair beyond discretisation error; (ii) a configuration is exhibited in which loading is even while $M \nmid n^d$, or a simultaneous return earlier than $\text{lcm}(p_1, p_2)$; (iii) any pair n_1, n_2 whose shared spectrum differs from that of Z/gcd , or coupled splittings departing from first/second order. Each check is one script run. The Tier II readings inherit the fates of the Tier I facts they cite.